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1. Background Research for the project

1.1 Context

Machine learning models often achieve high predictive performance, but their decision-making processes remain difficult to interpret. This lack of transparency limits trust, hinders model diagnosis, and reduces the effectiveness of machine learning systems in real-world applications. As a result, there is increasing interest in explainable artificial intelligence (XAI), which aims to provide insight into how models generate predictions.

Understanding model behaviour is essential not only for machine learning practitioners but also for stakeholders who rely on model outputs. The Making Visualization Scalable (MAVIS) project investigates how visualisation and interaction techniques can improve the understanding, diagnosis, and explanation of machine learning classification models. It highlights that visualisations, while useful, can become ineffective when applied to large or complex datasets due to issues such as overplotting, ineffective encoding, and interaction limitations.

Existing XAI techniques, including feature importance methods and local explanation approaches, provide insights into model predictions but are often presented through static or fragmented visualisations. These approaches typically focus on individual models and do not adequately support comparison across multiple models. As a result, it remains difficult to analyse how different models behave, where they agree or disagree, and how their predictions are influenced by different features.

Visual analytics offers a promising approach by combining interactive visualisation with analytical reasoning, enabling users to explore model behaviour dynamically. Systems such as ClaVis demonstrate how interactive interfaces can support the comparison of classification models and the exploration of model behaviour. However, such systems are generally dataset-agnostic and often focus on specific aspects of model development, rather than integrating explanation techniques with comparative analysis in a unified framework.

This project is developed in alignment with the MAVIS research direction, focusing on the use of interactive visualisation to support the understanding, comparison, and critical evaluation of classification models. In particular, it emphasises tasks such as model diagnosis, improvement, and selection, while addressing limitations in existing visual and explainable AI approaches.

1.2 Problem statement

Despite the availability of numerous explainable AI techniques and visual analytics systems, there remains a lack of integrated approaches that support the comprehensive understanding and comparison of machine learning classification models. Existing XAI methods typically provide isolated explanations, such as feature importance scores or local instance-level interpretations, but do not facilitate systematic comparison between models or support deeper reasoning about their behaviour.

Current visual analytics systems, including tools such as ClaVis, demonstrate the value of interactive visualisation for exploring model performance and behaviour. However, these systems often focus on specific aspects such as hyperparameter tuning or prediction comparison, and do not fully integrate explanation techniques with model comparison in a way that supports critical analysis. In particular, there is limited support for examining how explanations differ across models, how misclassifications arise, and how decision behaviour varies between linear and non-linear approaches.

Furthermore, visualisations themselves can introduce challenges. As highlighted in the MAVIS project, visual representations may become ineffective when dealing with complex datasets, leading to issues such as overplotting, misleading visual encodings, and cognitive overload. These limitations can result in misinterpretation of model behaviour and reduce the effectiveness of explanation techniques.

The core problem is the absence of an integrated system that enables users to explore, explain, and compare classification models in a structured and interactive manner. Existing approaches either provide isolated explanations or limited comparison capabilities, but do not support deeper reasoning about model behaviour, differences, and limitations.

1.3 Possible solution

To address the identified problem, this project proposes the design and implementation of an interactive visual analytics system that integrates model explanation and comparison within a unified framework. The system is designed to support key analytical tasks, including performance comparison, feature importance analysis, misclassification investigation, and examination of decision behaviour.

The system will use two classification models: Logistic Regression and Random Forest. These models are selected to represent fundamentally different decision behaviours, enabling comparison between linear and non-linear approaches. Logistic Regression provides a transparent baseline, while Random Forest captures more complex relationships within the data.

Two real-world tabular datasets will be used: the HELOC dataset and the Bank Marketing dataset. These datasets are selected to provide complementary analytical characteristics. The HELOC dataset introduces numerical complexity through encoded special values and domain-specific feature characteristics, making it suitable for analysing limitations of explanation methods. In contrast, the Bank Marketing dataset contains mixed feature types and a large number of instances, supporting visual exploration, interaction, and scalability analysis.

The system is intentionally designed to operate across datasets with different structural properties. While HELOC provides a homogeneous numerical setting, the Bank Marketing dataset introduces heterogeneous feature types, enabling evaluation of the system under varied data conditions.

The proposed system will include interactive visualisations that allow users to:

- compare model performance metrics

- explore global and local feature importance
- identify and analyse misclassified instances
- examine differences in model decision behaviour

Interactivity is central to the design, enabling users to dynamically explore data, switch between models, and focus on specific subsets of interest. The system supports analytical reasoning rather than static presentation, allowing users to investigate how and why models behave differently.

This combination enables evaluation of the system across both homogeneous and heterogeneous data conditions, strengthening the robustness of analysis.

1.4 How to demonstrate the quality of the solution

The quality of the proposed system will be evaluated through analytical and comparative assessment rather than user studies. The evaluation focuses on how effectively the system supports model understanding, comparison, and critical analysis.

Model performance will first be assessed using standard metrics such as accuracy, establishing a baseline for comparison between Logistic Regression and Random Forest. The purpose is to determine whether the models exhibit different predictive behaviours across the selected datasets.

The system will then be evaluated based on its ability to reveal meaningful insights through visualisation. Feature importance analysis will be used to examine how different models prioritise features and whether these patterns are consistent or vary across models.

Misclassification analysis will assess how effectively the system supports model diagnosis. By identifying incorrectly classified instances, the system should enable investigation of error patterns and difficult cases.

Decision behaviour will be examined by analysing how models separate classes and how their predictions differ. This is particularly relevant when comparing linear and non-linear models.

Finally, a critical evaluation will identify limitations of the system and its visualisations. This includes examining misleading representations, the impact of data complexity, and trade-offs between interpretability and predictive performance.

2. Scope for this project

2.1 Aim

The aim of this project is to design, implement, and evaluate an interactive visual analytics system for exploring, explaining, and comparing classification models, with a focus on supporting model diagnosis, improvement, and selection through key explainable AI tasks.

2.2 Objectives

The project will achieve the following objectives:

1. Conduct a structured literature review of explainable AI methods and visual analytics approaches for classification models, identifying key limitations and challenges.
2. Select and preprocess suitable tabular datasets including handling of missing values, encoding of categorical features, and validation of data quality.
3. Implement and evaluate classification models using Logistic Regression and Random Forest, and assess their performance using standard metrics such as accuracy, precision, recall, and F1-score.
4. Design and develop interactive visualisations that support key XAI tasks, including feature importance, misclassification analysis, class separation, and decision behaviour.
5. Develop an interactive visual analytics system that enables users to switch between models, explore explanations, and compare model behaviour dynamically.
6. Evaluate the system through analytical case studies by comparing model behaviour, analysing explanation differences, and critically assessing limitations and potential improvements.

2.3 Deliverables

- Interactive visual analytics system
- Implementation code and GitHub repository
- Visualisation components supporting key analytical tasks
- Analytical evaluation and findings
- Final MSc dissertation report

3. Project schedule

3.1 Methodology

This project follows a research-driven, iterative development methodology that integrates data science workflows with visual analytics system design. The methodology is structured to support progressive development, evaluation, and refinement of the system.

The project begins with a literature review and problem analysis, focusing on explainable AI techniques and visual analytics systems. This stage establishes the research context, identifies limitations in existing approaches, and informs system design decisions.

The next stage involves data preparation and modelling. Selected datasets are preprocessed through handling of missing or encoded values, encoding of categorical features where required, and feature scaling if necessary. Logistic Regression and Random Forest models are then trained to provide contrasting behaviours for analysis.

Following this, explanation methods are implemented to generate both global and local interpretations of model behaviour. Misclassified instances are identified to support error analysis and model diagnosis.

The system is then developed as an interactive visual analytics platform. Model outputs and explanations are integrated into visual components that allow users to explore performance, feature importance, misclassifications, and decision behaviour through interaction.

Finally, the system is evaluated through analytical and comparative assessment. The evaluation focuses on differences between models, insights derived from visualisations, and the identification of limitations. This ensures that the system is assessed not only for functionality but also for its ability to support meaningful reasoning.

3.2 Tasks, milestones and timeline

The project is structured into the following key tasks and milestones:

Phase 1: Background Research and Planning

- Literature review on XAI and visual analytics systems
 - Analysis of ClaVis and MAVIS frameworks
 - Finalisation of project scope and datasets
- Milestone:** Completion of S&P document

Phase 2: Data Preparation and Exploration

- Data cleaning and preprocessing (HELOC and Bank Marketing datasets)
 - Handling missing values and encoding features
 - Exploratory data analysis
- Milestone:** Clean and prepared datasets

Phase 3: Model Development

- Implementation of Logistic Regression and Random Forest
 - Model training and validation
 - Performance evaluation (accuracy and related metrics)
- Milestone:** Trained and evaluated models

Phase 4: Explanation and Analysis

- Implementation of feature importance methods
 - Identification and analysis of misclassified instances
 - Initial comparison of model behaviour
- Milestone:** Explanation outputs and analysis results

Phase 5: System Development

- Design and implementation of interactive visualisations

- Integration of models and explanation components
 - Development of user interface for interaction
- Milestone:** Functional interactive system

Phase 6: Evaluation and Reporting

- Comparative analysis of models using visualisations
 - Identification of insights and limitations
 - Final report writing and documentation
- Milestone:** Completed dissertation and system submission

3.3 Risk assessment (if appropriate)

Several risks have been identified for this project, along with mitigation strategies:

1. Data-related challenges

- **Risk:** Issues with data quality, missing values (especially in HELOC), or preprocessing complexity
- **Mitigation:** Use established preprocessing techniques (imputation, encoding) and validate data before modelling

2. Model performance limitations

- **Risk:** Models may not show significant differences, limiting comparative analysis
- **Mitigation:** Use datasets with sufficient complexity and ensure proper feature engineering and evaluation

3. Complexity of visualisation design

- **Risk:** Visualisations may become cluttered or difficult to interpret
- **Mitigation:** Focus on clear, task-driven visualisations and iterative refinement based on usability

4. Time constraints

- **Risk:** Delays in implementation or integration may affect later stages
- **Mitigation:** Follow structured milestones and prioritise core functionality before advanced features

5. Technical integration challenges

- **Risk:** Difficulty integrating models, explanations, and visual interface
- **Mitigation:** Develop and test components incrementally before full integration

References

- Ruddle, R. A., et al. (2022). *Making Visualization Scalable (MAVIS): Supporting the Analysis of Machine Learning Models*. University of Leeds.
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why Should I Trust You?" *Explaining the Predictions of Any Classifier*. Proceedings of the 22nd ACM SIGKDD International Conference.
- Lundberg, S. M., & Lee, S.-I. (2017). *A Unified Approach to Interpreting Model Predictions*. Advances in Neural Information Processing Systems.
- Yeh, I. C., & Lien, C. H. (2009). *The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients*. Expert Systems with Applications.
- UCI Machine Learning Repository (2014). *Bank Marketing Dataset*. [online]
- FICO (2018). *Explainable Machine Learning Challenge (HELOC Dataset)*.

Appendix A. How ethical issues are addressed

This project uses publicly available datasets that do not contain personally identifiable information. The datasets are anonymised and widely used for research and educational purposes. As such, there are no direct ethical concerns related to data privacy or participant consent.

However, the project involves analysing models used in decision-making contexts such as financial risk assessment. Therefore, care will be taken when interpreting and presenting results to avoid misleading conclusions. In particular, the limitations of models and explanation methods will be explicitly discussed.

The system will be developed and evaluated solely for academic purposes, and no real-world decisions will be made based on its outputs. All work will comply with the University of Leeds ethical guidelines for computing projects.